

EPREUVE DU 20 JANVIER 2020

Let $(\Omega, \mathcal{F}, P)$ be a complete probability space. Let $T > 0$. Let (W_t) be a standard Brownian motion with canonical filtration $(\mathcal{F}_t)$.

1. Let $(\sigma_s), (\beta_s)$ be two progressively measurable bounded processes. and a continuous progressively measurable process (γ_t) . Let $\xi_0 \in \mathbb{R}$. We set

$$R_t = \exp \left(\int_0^t \sigma_s dW_s + \int_0^t \left(\beta_s - \frac{\sigma_s^2}{2} \right) ds \right), t \in [0, T]. \quad (1)$$

- (a) Show that R is a solution to the SDE

$$dR_t = R_t \sigma_t dW_t + R_t \beta_t dt.$$

- (b) We set $\rho_t := \frac{1}{R_t}$. Determine a similar linear SDE solved by ρ_t .

- (c) Let (ξ_t) be a solution to the linear SDE

$$\xi_t = \xi_0 + \int_0^t \xi_s \sigma_s dW_s + \int_0^t \xi_s \beta_s ds + \int_0^t \gamma_s ds. \quad (2)$$

Show that

$$\xi_t \rho_t = \xi_0 + \int_0^t \rho_s \gamma_s ds, t \in [0, T].$$

2. Let $a, b, \mu : \mathbb{R} \rightarrow \mathbb{R}$ Lipschitz functions. Let $x_0, y_0 \in \mathbb{R}$. Let X and Y be stochastic processes solving the SDEs

$$\begin{aligned} X_t &= x_0 + \int_0^t a(X_s) dW_s + \int_0^t b(X_s) ds, \\ Y_t &= y_0 + \int_0^t a(Y_s) dW_s + \int_0^t \mu(Y_s) ds. \end{aligned} \quad (3)$$

We set $\xi_t = X_t - Y_t, t \in [0, T]$. We define here

$$\beta_t = \begin{cases} \frac{b(X_t) - b(Y_t)}{X_t - Y_t} & : X_t \neq Y_t \\ 0 & : X_t = Y_t, \end{cases}$$

$$\sigma_t = \begin{cases} \frac{a(X_t) - a(Y_t)}{X_t - Y_t} & : X_t \neq Y_t \\ 0 & : X_t = Y_t. \end{cases}$$

- (a) Show that (ξ_t) solves one equation of type (2) specifying ξ_0 and γ .
 - (b) We suppose $x_0 \geq y_0$ and $b(x) \geq \mu(x), \forall x \geq 0$. Conclude that a.s. $X_t \geq Y_t, t \in [0, T]$.
3. Let X still be the process solving (3). Let $g : \mathbb{R} \rightarrow \mathbb{R}$ with linear growth.
- (a) Let $M_t := E(g(X_T) | \mathcal{F}_t), t \in [0, T]$. Show that M is a square integrable martingale. Explain why M has a continuous modification.
 - (b) Show the existence of $v : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ such that $M_t = v(t, X_t), t \in [0, T]$.
 - (c) Let $v : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ supposed to be of class $C^{1,2}$. Show that

$$M_t = v(0, x_0) + \int_0^t (a \partial_x v)(s, X_s) dW_s, t \in [0, T].$$

- (d) Show that $E(g(X_T)) = v(0, x_0)$.
4. Let X still be the process solving (3).
- (a) Recall the meaning for a probability Q on $(\Omega, \mathcal{F})$ to be equivalent to P .
 - (b) Suppose $a \neq 0$ and that the function $\frac{b}{a}$ is bounded. We suppose $\mathcal{F} = \mathcal{F}_T$. Show the existence of an equivalent probability Q on $(\Omega, \mathcal{F})$ such that X is an $(\mathcal{F}_t)$ -local martingale.
 - (c) Justify finally that X is (under Q) even a square integrable martingale.

5. (a) Let $(G_t)_{t \in [0, T]}$ be a mean-zero continuous Gaussian process and we define

$$c(s, t) = \text{Cov}(G_s, G_t), (s, t) \in [0, T]^2.$$

Show that $\int_0^T G_s ds$ is a mean-zero Gaussian r.v.

- (b) Show that the variance of previous Gaussian r.v. equals

$$2 \int_0^T \int_0^{s_1} c(s_1, s_2) ds_2 ds_1.$$

- (c) We set $G_t = W_t, t \in [0, T]$, where W is a standard Brownian motion. Deduce the law of $\int_0^T G_s ds$, its expectation and variance.

Document autorisé. Deux pages (ou une feuille) de résumé.

Elements of Stochastic Calculus-First Part (FQ307-OMI302).

END